
PACE: Marrying generalization in PArameter-efficient fine-tuning with Consistency rEgularization

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Abstract

Parameter-Efficient Fine-Tuning (PEFT) effectively adapts pre-trained transformers to downstream tasks. However, the optimization of tasks performance often comes at the cost of generalizability in fine-tuned models. To address this issue, we theoretically connect smaller weight gradient norms during training and larger datasets to the improvements in model generalization. Motivated by this connection, we propose reducing gradient norms for enhanced generalization and aligning fine-tuned model with the pre-trained counterpart to retain knowledge from large-scale pre-training data. Yet, naive alignment does not guarantee gradient reduction and can potentially cause gradient explosion, complicating efforts to manage gradients. To address such an issue, we propose PACE, marrying generalization of PArameter-efficient fine-tuning with Consistency rEgularization. We perturb features learned from the adapter with the multiplicative noise and ensure the fine-tuned model remains consistent for same sample under different perturbations. Theoretical analysis shows that PACE not only implicitly regularizes gradients for enhanced generalization, but also implicitly aligns the fine-tuned and pre-trained models to retain knowledge. Experimental evidence supports our theories. PACE surpasses existing PEFT methods in visual adaptation tasks (VTAB-1k, FGVC, few-shot learning, domain adaptation) showcasing its potential for resource-efficient fine-tuning. It also improves LoRA in text classification (GLUE) and mathematical reasoning (GSM-8K). The code is available at github.com/MaxwellYaoNi/PACE.

1 Introduction

Transformers [68], with the self-attention mechanism [3] capturing long-range dependencies in data, succeed in various deep learning tasks, including image classification (ViT [16]), multimodal learning (CLIP [55]), image synthesis (StableDiffusion [57]), semantic segmentation (SAM [33]) and text generation (LLaMA [65]). The success of transformers can be largely attributed to the availability of abundant data, such as ImageNet [11] and Laion5B [60], which empower researchers to scale up these models by training them under an enormous number of parameters.

Such huge models, with knowledge from large-scale pre-training [63], constitute on foundation models that can be easily adapted to various downstream tasks through full fine-tuning or linear probing [20], eliminating the need for task-specific model design [8]. However, full fine-tuning is storage-intensive and infeasible for maintaining separate model weights as the number of tasks grows, while linear probing, which only trains the last head layer, yields inferior adaptation performance.

To overcome these limitations, Parameter-Efficient Fine-Tuning (PEFT) [24] fine-tunes only a small subset of parameters, thereby reducing storage requirements while surpassing the performance of

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full fine-tuning and linear probing. These advantages have popularized PEFT and inspired the development of various PEFT methods for deep learning tasks, which can be categorized into two groups: those increasing inference cost and cost-efficient ones. The first group introduces additional learning branches, such as non-linear adapters [25, 8], or concatenates learnable parameters with input tokens, *e.g.*, visual prompts [28, 82, 52], increasing inference cost. The second group, focuses on cost-efficiency by lower-rank adaptation in linear layers [7, 26], or affine transformations such as SSF [41] and RepAdapters [45], which can be reparameterized during inference for efficiency.

Despite the superiority and efficiency of PEFT, prioritizing optimization for downstream tasks compromises the generalizability of fine-tuned models, yielding suboptimal performance. Although some analyses have been conducted on PEFT [63, 27, 18, 72, 39], they fail to fully explain the generalization of PEFT, leading to ineffective strategies for improving generalization.

To address this gap in understanding generalization in PEFT, we establish a theoretical connection from generalization theory: smaller weight gradient norms and larger data volumes contribute to better generalization. Motivated by this, we propose reducing weight gradient norms and aligning output space of the fine-tuned model with the pre-trained one to retain knowledge captured from large pre-training data. Yet, theoretical analyses reveal this naive alignment does not guarantee gradient regularization and can even cause gradient explosion, complicating efforts for gradient management. To address this issue, we propose perturbing features learned from the adapter with multiplicative noise and constraining the network output to be consistent across different perturbations.

Our method, called PACE, marries generalization of PARAMeter-efficient fine-tuning with Consistency rEGularization. Its name, PACE, reflects our goal of keeping the output behavior of the fine-tuned model *in pace with* the pre-trained one. Despite its simplicity, theoretical analysis confirms that PACE not only implicitly regularizes weight gradients for better generalization but also implicitly aligns the fine-tuned model with the pre-trained counterpart to retain knowledge from large-scale pre-training data. Experimental evidence supports our theories. PACE improves existing PEFT methods, achieving superior results across six adaptation benchmarks. Our key contributions are:

- i. We establish a theory connecting smaller weight gradient norms and larger datasets with enhanced generalization, motivating gradient reduction and model alignment for fine-tuning.
- ii. We propose PACE, a simple yet effective method perturbing features from adapters with multiplicative noise and constraining output of fine-tuned model to be consistent across perturbations.
- iii. Our theoretical and empirical evidence confirms that PACE implicitly regularizes gradients and aligns the fine-tuned model with the pre-trained one. PACE excels on 4 visual adaptation tasks.
- iv. We provide novel theoretical explanations of how gradient penalization and consistency regularization benefit generalization, offering fundamental insights applicable across deep learning.

2 Related work

Parameter-Efficient Fine-Tuning (PEFT). LoRA [26] uses low-rank decomposition to reduce parameters and treats adapters as side paths. SSF [41] proposes affine transformations on latent features. FacT [30] decomposes and reassembles parameter matrices in ViT. Surgical fine-tuning [36] of different network parts improves adaptation to distribution shifts. FLoRA [74] performs a batched low-rank adaptation. GLoRA [7] unifies cost-efficient PEFT methods. NOAH [82] uses parameter search on neural prompts. ARC [14] leverages cross-layer ViT similarity, parameter-sharing adapter and scaling factors for lower fine-tuning cost. RLRR [15] incorporates a residual term for flexibility while preserving pre-trained representation. RepAdapter [45] reparameterizes adapters for efficient inference. Res-tuning [29] unbinds tuners from the backbone for memory efficiency. Zhao *et al.* [84] show impressive fine-tuning results by tuning layernorm in attention. OFT [54] and BOFT [42] propose orthogonal fine-tuning to preserve hypersphere energy between neurons.

Consistency Regularization. Fixmatch [61] applies consistency regularization over augmented images for semi-supervised learning. Openmatch [59] utilizes it on outlier predictions for open-set semi-supervised learning. R-Drop [76] applies it to transformers [68] with dropout for NLP tasks. CR [79] applies it over augmented real and fake images for GAN training. CAGAN [50] enforces consistency on discriminators with dropout for GAN training. Despite the empirical success of consistency regularization demonstrated by previous works, theoretical analysis is lacking. While NICE [47] demonstrates that consistency regularization lowers latent feature gradients for stable

GAN training, it fails to reveal reduced weight gradient for enhanced generalization. Our study goes beyond prior works by providing a theoretical link between smaller weight gradients and improved generalization, effectively marrying generalization of PEFT with consistency regularization.

Generalization of Fine-Tuning. Li *et al.* [38] constrain the fine-tuned model’s closeness to the pre-trained model in weight space. Fu *et al.* [18] induce sparsity on PEFT for better generalization. Wang *et al.* [72] studies generalization of PEFT fine-tuning graph neural network. Zhang *et al.* [83] employ rank-1 gradient boosting (GB) updates supported by the GB theoretical framework. VioLET [73], PromptSRC [31] and CoPrompt [58] naively align the fine-tuned model with the pre-trained one for enhanced generalization or avoiding forgetting. Additionally, L2SP [77], DELTA [40], and FTP [64] aim to retain pre-trained knowledge by aligning fine-tuned models with pre-trained ones, reducing distance in weight space, feature space and using projected gradient descent, respectively. However, they fail to provide a theoretical analysis for this alignment. Our study goes beyond understanding generalization of PEFT by discovering the benefits of gradient regularization and model alignment. We propose PACE to match both requirements, paving a comprehensive understanding for PEFT.

Gradient regularization. Previous studies have empirically shown that gradient regularization improves performance [67, 85, 48, 49] and adversarially robust accuracy [13]. However, they lack theoretical connection between smaller gradient norms and better generalization [17, 81, 6]. We bridge this gap by establishing a fundamental theory between reduced gradient norms and improved generalization, providing a solid foundation for future research on enhancing generalization.

3 Approach

We begin with a unified perspective on cost-efficient PEFT based on GLoRA [7], linking generalization with gradients and large-scale data, and motivating the alignment of the fine-tuned model with the pre-trained model to leverage its knowledge. We identify limitations of naive alignment in gradient regularization and introduce PACE, which implicitly enhances gradient regularization and model alignment. We conclude with theoretical justification and efficient implementations.

3.1 A unified perspective on cost-efficient PEFT methods

The transformer architectures [68, 16] have excelled in natural language processing and computer vision tasks through their powerful sequential modeling capabilities. This success stems from their ability to process text/image tokens through L transformer blocks, where each block contains self-attention and MLP modules primarily composed of linear layers. These linear layers enable the self-attention mechanism to capture long-range dependencies, allowing transformers to achieve superior performance when scaled to a huge number of parameters and trained on extensive datasets.

With massive parameters, pre-trained on large-scale data, transformers serve as foundation models that can be fine-tuned for downstream tasks using limited data. However, fully fine-tuning all parameters for various downstream tasks requires substantial memory and can lead the forgetting of pre-trained knowledge. To alleviate this without increasing inference cost, adapters with lightweight parameters are often preferred for fine-tuning. Let $\bar{h}_0(\cdot)$ be a transformation branch within the pre-trained transformer. Current adapters can be unified as introducing a residual branch $\Delta\bar{h}$ to form a new transformation $\bar{h}$:

$$\bar{h}(\mathbf{a}) = \bar{h}_0(\mathbf{a}) + \Delta\bar{h}(\mathbf{a}). \quad (1)$$

Here, $\mathbf{a}$ is the input and $\bar{h}_0(\cdot)$ can represent MLP modules, as in Adapter [25] and AdaptFormer [8], or linear layers in self-attention and MLP modules, as in [26, 7, 12, 34]. In SSF [41], $\bar{h}_0(\cdot)$ is the identity mapping and $\Delta\bar{h}(\mathbf{a}) = \mathbf{a} \odot (\gamma - \mathbf{1}) + \beta$ with γ and β as affine transformation parameters.

Given that linear layers are key components in transformer, tuning them offers a flexible and effective way to adapt models to downstream tasks. This work focuses on methods that tune the linear layer without increasing inference cost. Let $(\mathbf{W}_0, \mathbf{b}_0)$, $(\Delta\mathbf{W}, \Delta\mathbf{b})$, and $(\mathbf{W}, \mathbf{b})$ be the parameters of pre-trained model, adapter and fine-tuned model, respectively, where $\mathbf{W}_0, \Delta\mathbf{W}, \mathbf{W} \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ and $\mathbf{b}_0, \Delta\mathbf{b}, \mathbf{b} \in \mathbb{R}^{d_{\text{out}}}$. Fine-tuning a linear layer in self-attention or MLP module can be formed as:

$$\begin{aligned} h(\mathbf{a}) &= \mathbf{W}\mathbf{a} + \mathbf{b} = (\mathbf{W}_0 + \Delta\mathbf{W})\mathbf{a} + (\mathbf{b}_0 + \Delta\mathbf{b}) \\ &= h_0(\mathbf{a}) + \Delta h(\mathbf{a}) = (\mathbf{W}_0\mathbf{a} + \mathbf{b}_0) + (\Delta\mathbf{W}\mathbf{a} + \Delta\mathbf{b}). \end{aligned} \quad (2)$$

Based on GLoRA [7], cost-efficient PEFT methods for linear layers vary in the form of $\Delta\mathbf{W}, \Delta\mathbf{b}$:

LoRA_{add}: $\Delta W = W_d W_u$, $\Delta b = b_{\text{loa}}$ where $W_d \in \mathbb{R}^{d_{\text{out}} \times r}$, $W_u \in \mathbb{R}^{r \times d_{\text{in}}}$, and r is the rank.

LoRA_{mul}: $\Delta W = W_0 \odot (W_d W_u)$, $\Delta b = b_0 \odot b_{\text{loa}}$, including RepAdapter [45] via reparameterization.

VPT_{add}: ΔW is zero, $\Delta b = W_0 P$, with learnable $P \in \mathbb{R}^{d_{\text{in}} \times 1}$ as layer-wise visual prompt. We use VPT_{add} to differentiate from VPT [28], which concatenates P with tokens, increasing inference cost.

3.2 Generalization of deep neural networks

Having established a unified perspective on cost-efficient PEFT, we now motivate our method from a perspective on improving generalization of neural networks to enhance performance on unseen data. Consider a network $f := \phi(g(x))$ with l layers, where g is feature extractor and ϕ is the classification head. Let $\theta := \{(W^{(i)}, b^{(i)})\}_{i=1}^l$ be the parameter set with dimension d and $\mathcal{D}^n := \{(x_i, y_i)\}_{i=1}^n$ be the training set of size n drawn *i.i.d.* from distribution $\mathcal{D}$, which contains infinite data. The following lemma from [17] explains the relationship between the empirical and population loss.

Lemma 1 (Theorem 1 from [17]) *Let $\mathcal{L}_{\mathcal{D}^n}(\theta)$ be the empirical loss function over f on training set $\mathcal{D}^n$ and $\mathcal{L}_{\mathcal{D}}(\theta)$ be the population loss. For any $\rho > 0$, with high probability over $\mathcal{D}^n \sim \mathcal{D}$, we have*

$$\mathcal{L}_{\mathcal{D}}(\theta) \leq \max_{\|\epsilon\|_2 \leq \rho} \mathcal{L}_{\mathcal{D}^n}(\theta + \epsilon) + R\left(\frac{\|\theta\|_2^2}{\rho^2}, \frac{1}{n}\right), \quad (3)$$

where $R : (\mathbb{R}_+, \mathbb{R}_+) \rightarrow \mathbb{R}_+$ is an increasing function (under conditions on $\mathcal{L}_{\mathcal{D}}(\theta)$ and n as in §B.5).

Lemma 1 bounds the population loss by the empirical loss with perturbed weights, indicating that a minimal empirical loss increase from small weight perturbations implies low population loss.

By observing that the maximum of $\mathcal{L}_{\mathcal{D}^n}$ is achieved at $\epsilon = \frac{\rho \nabla_{\theta}}{\|\nabla_{\theta}\|_2}$, where ∇_{θ} is the gradient of $\mathcal{L}_{\mathcal{D}^n}$ at θ , and performing a Taylor expansion of $\mathcal{L}_{\mathcal{D}^n}$ around θ , we formulate the following theorem.

Theorem 1 *Denote ∇_{θ} as the gradient and $\lambda_{\max}^H$ as the largest eigenvalue of the Hessian matrix H_{θ} of $\mathcal{L}_{\mathcal{D}^n}$ at θ . For any $\rho > 0$, with high probability over training set $\mathcal{D}^n \sim \mathcal{D}$, we have*

$$\mathcal{L}_{\mathcal{D}}(\theta) \leq \mathcal{L}_{\mathcal{D}^n}(\theta) + \rho \|\nabla_{\theta}\|_2 + \frac{\rho^2}{2} \lambda_{\max}^H + R\left(\frac{\|\theta\|_2^2}{\rho^2}, \frac{1}{n}\right). \quad (4)$$

Here, higher-order terms from the Taylor expansion are incorporated into $R\left(\frac{\|\theta\|_2^2}{\rho^2}, \frac{1}{n}\right)$, which is related to weights norm and inversely related to the training data size n .

Theorem 1 (proof in §B.1) outlines strategies for enhancing generalization. They involve regularizing weight norms and the largest Hessian eigenvalues, and crucially, increasing data size n and reducing the weight gradient norms (illustrated in Figure 1). However, excessive reduction should be avoided as it could impair network’s representation capacity, yielding higher empirical and population loss.

3.3 Motivation and limitation of aligning the fine-tuned model with the pre-trained model

Theorem 1 emphasizes that large-scale data and smaller gradient magnitudes are essential for better generalization in neural network training. Therefore, aligning the fine-tuned model with the pre-trained one is crucial, as it ensures retention of knowledge obtained from large-scale data, preserving generalization. PEFT methods, often outperforming full fine-tuning, achieve this alignment by limiting the number of trainable parameters, restricting the model’s capacity to deviate from the pre-trained one. However, the training objective prioritizes downstream task performance, compromising alignment with pre-trained knowledge. While sparsity regularization [18] and weight decay on adapter weights help, they do not ensure alignment, as even small weight changes can lead to significant divergence in output space [75, 21, 17]. Therefore, we propose to achieve the alignment by reducing the FP-distance (output distance between fine-tuned and pre-trained models on training samples):

$$D^{\text{fp}}(\theta) = \frac{1}{n} \sum_{i=1}^n \|f(x_i; \theta) - f(x_i; \theta_0)\|_2^2, \quad \theta = \theta_0 + \Delta\theta, \quad (5)$$

where $\theta, \theta_0, \Delta\theta \in \mathbb{R}^d$ are parameters for the fine-tuned model, pre-trained model and the adapter.

While reducing FP-distance keeps the fine-tuned model close to the pre-trained model, thus preserving its knowledge, it does not ensure reduced gradient magnitudes, leading to suboptimal generalization.

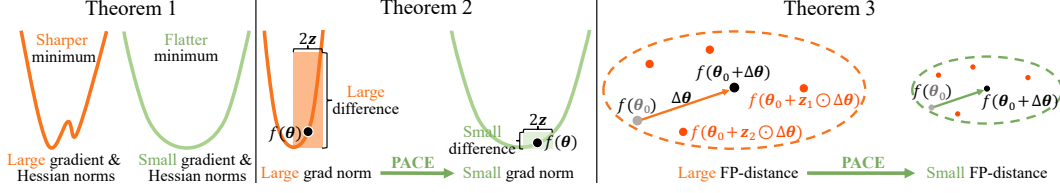


Figure 1: Thm. 1: A flatter minimum has smaller gradient and Hessian norms, yielding better generalization. Thm. 2: Large gradient norms indicate large differences among perturbations. PACE minimizes these differences, reducing gradient norms. Thm. 3: Minimizing all pairs of distances between $f(\theta_0 + z_1 \odot \Delta\theta)$ and $f(\theta_0 + z_2 \odot \Delta\theta)$ where $z_1, z_2 \sim \mathcal{N}(\mathbf{1}, \sigma^2 \mathbf{I})$ also reduces FP-distance (between fine-tuned $f(\theta_0 + \Delta\theta)$ and pre-trained $f(\theta_0)$), especially when $z_1 = \mathbf{1}$, $z_2 = \mathbf{0}$ or vice versa.

To understand the gradient-related limitations in this alignment, we assume $\Delta\theta$ is small enough for a Taylor expansion approximation. Following standard practices [17, 80, 2], we perform the expansion up to the second-order terms. Given the independence between elements in squared L_2 distances (§B.4) and to simplify our theories, we analyze a one-dimensional output for a single *i.i.d.* sample, which leads us to the following proposition.

Proposition 1 Assuming $\Delta\theta$ is small, denote $f(\theta) \in \mathbb{R}$ as the one-dimensional output for x , with ∇ and H as its gradient and Hessian at θ . FP-distance over x can be decomposed as follows:

$$\begin{aligned} [f(\theta) - f(\theta_0)]^2 &= [f(\theta) - f(\theta - \Delta\theta)]^2 \approx [f(\theta) - [f(\theta) - \Delta\theta^T \nabla + \frac{1}{2} \Delta\theta^T H \Delta\theta]]^2 \\ &\approx [\Delta\theta^T \nabla - \frac{1}{2} \Delta\theta^T H \Delta\theta]^2. \end{aligned} \quad (6)$$

Prop. 1 establishes the relationship between weight gradients, adapter weights, and FP-distance. However, it remains unclear if it regulates gradients. Our experiments show that minimizing FP-distance can sometimes increase gradient magnitude, complicating efforts for managing gradient.

3.4 Consistency regularization

To achieve better generalization by both regularizing gradients and aligning the fine-tuned model with the pre-trained model, we propose a consistency regularization loss for f , encouraging invariance of f to the same input under varying multiplicative noise perturbations on the adapter weights, as follows:

$$D^{\text{pace}}(\theta) = \frac{1}{n} \sum_{i=1}^n \mathbb{E}_{z_1, z_2} \|f(x_i; \theta_0 + z_1 \odot \Delta\theta) - f(x_i; \theta_0 + z_2 \odot \Delta\theta)\|_2^2, \quad (7)$$

where $z_1, z_2 \sim \mathcal{N}(\mathbf{1}, \sigma^2 \mathbf{I})$ is the multiplicative noise applied on adapter weight. To understand the generalization benefits in this consistency regularization, we simplify the analysis by focusing on one-dimensional output for a single sample, resulting in the following theorem.

Theorem 2 Using notations from Prop. 1, let $f(\theta_0 + z \odot \Delta\theta) \in \mathbb{R}$ be the one-dimensional output for x . Define $\Delta\theta_j$ as j -th element in $\Delta\theta$, ∇_j as the j -th element in ∇ and H_{jk} as the (j, k) -entry in H . With $z_1, z_2 \sim \mathcal{N}(\mathbf{1}, \sigma^2 \mathbf{I})$, the consistency loss over x can be approximated as:

$$\begin{aligned} &\mathbb{E}_{z_1, z_2} [f(\theta_0 + z_1 \odot \Delta\theta) - f(\theta_0 + z_2 \odot \Delta\theta)]^2 \\ &\approx 2\sigma^2 \sum_j \Delta\theta_j^2 \nabla_j^2 + \sigma^4 \sum_{j,k} \Delta\theta_k^2 \Delta\theta_j^2 H_{jk}^2 = 2\sigma^2 \|\Delta\theta \odot \nabla\|_2^2 + \sigma^4 \|(\Delta\theta \Delta\theta^T) \odot H\|_F^2. \end{aligned} \quad (8)$$

Theorem 2 (proof in §B.2) shows that the consistency regularization essentially penalizes the first- and second-order gradients of f at θ (illustrated in Figure 1), with the regularization strength controlled by the noise variance σ^2 and adaptively influenced by the magnitude of elements in adapter weight $\Delta\theta$. Thus, minimizing the consistency loss implicitly regularizes the gradients, improving generalization.

With the FP-distance in Prop. 1 and consistency loss in Theorem 2, we establish their relationship as:

Theorem 3 With d as the dimension of θ , Eq. 6 can be upper-bounded as:

$$[\Delta\theta^T \nabla - \frac{1}{2} \Delta\theta^T H \Delta\theta]^2 \leq 2d \|\Delta\theta \odot \nabla\|_2^2 + d^2 \|(\Delta\theta \Delta\theta^T) \odot H\|_F^2. \quad (9)$$

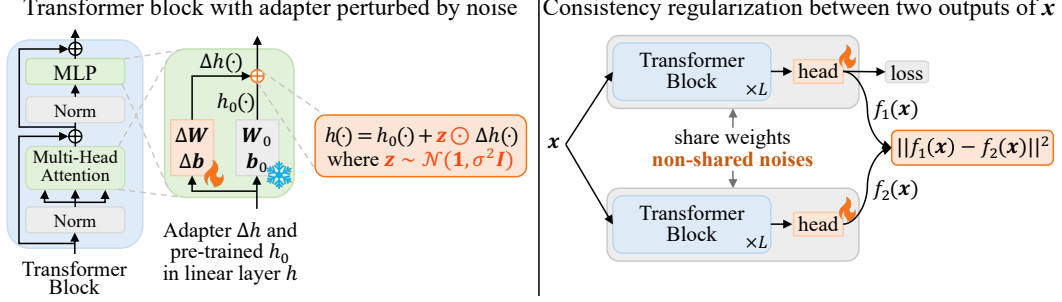


Figure 2: Our pipeline. Adapter $\Delta h(\cdot)$ and $h_0(\cdot)$ from pre-trained model form the linear layer h of Multi-Head Attention and MLP in fine-tuned model. We perturb $\Delta h(\cdot)$ with multiplicative noise and ensure the network remains consistent to same inputs under varying perturbations.

Theorem 3 (proof in B.3) establishes the relationship between Eq. 6 and Eq. 8, showing Eq. 6 is upper-bounded by terms involving $\|\Delta\theta \odot \nabla\|_2^2$ and $\|(\Delta\theta \Delta\theta^T) \odot H\|_F^2$ which appear in Eq. 8. Reducing these terms results in a decrease in Eq. 6. Thus minimizing the consistency loss implicitly aligns the fine-tuned and pre-trained models (illustrated in Figure 1), preserving pre-trained knowledge.

3.5 Efficient implementation of PACE

Providing different weight perturbations for each input in a mini-batch increases memory and computational demands. To avoid this, we perturb feature outputs from the adapter $\Delta h(\cdot)$, effectively simulating perturbation that shares noise across each row in the weight matrix ΔW . Our simple pipeline is shown in Figure 2. Consider $\mathbf{X} \in \mathbb{R}^{B \times T \times d_{in}}$ as a batch of data where B and T are the batch and token sizes. The calculation for the linear layer of the fine-tuned model, which utilizes pre-trained weights $\mathbf{W}_0, \mathbf{b}_0$ and adapter weights $\Delta \mathbf{W}, \Delta \mathbf{b}$, processes an output size of d_{out} as:

$$h_0(\mathbf{X}) = \mathbf{W}_0 \mathbf{X} + \mathbf{b}_0; \quad \Delta h(\mathbf{X}) = \Delta \mathbf{W} \mathbf{X} + \Delta \mathbf{b}, \quad (10)$$

$$h(\mathbf{X}) = h_0(\mathbf{X}) + \mathbf{Z} \odot \Delta h(\mathbf{X}). \quad (11)$$

Operator $\odot$ is the element-wise multiplication after expanding the left matrix $\mathbf{Z} \in \mathbb{R}^{B \times d_{out}} \sim \mathcal{N}(\mathbf{1}, \sigma^2 \mathbf{I})$ into $B \times T \times d_{out}$ where tokens within the same example share the same noise. Motivated by [37], the σ decreases linearly as block depth increases. Let $f_1(\cdot)$ and $f_2(\cdot)$ be two networks share same weights but do not share the noise patterns. The loss function for PACE is:

$$\mathcal{L}^{\text{PACE}} = \frac{1}{n} \sum_{i=1}^n \ell(f_1(\mathbf{x}_i), \mathbf{y}_i) + \lambda \|f_1(\mathbf{x}_i) - f_2(\mathbf{x}_i)\|_2^2, \quad (12)$$

where ℓ is the classification loss and λ is a hyperparameter controlling regularization strength. During inference, noise and regularization are omitted, $\Delta \mathbf{W}, \Delta \mathbf{b}$ are integrated with $\mathbf{W}_0, \mathbf{b}_0$ for efficiency:

$$\mathbf{W} = \mathbf{W}_0 + \Delta \mathbf{W}; \quad \mathbf{b} = \mathbf{b}_0 + \Delta \mathbf{b}; \quad h(\mathbf{X}) = \mathbf{W} \mathbf{X} + \mathbf{b}. \quad (13)$$

Efficient PACE variants. In §C, we present two variants that match the computational/memory costs of the baseline while achieving superior performance with substantially reduced resources.

4 Experiments

We combine LoRA_{mul} and VPT_{add} to form a strong baseline $\text{LoRA}_{\text{mul}} + \text{VPT}_{\text{add}}$, outperforming other combinations in most cases. We evaluate our method across four visual classification adaptation tasks: VTAB-1K [78], few-shot learning [30], FGVC [28] and domain adaptation [82]. We demonstrate PACE improves LoRA on GLUE [70] for text classification and GSM-8K [9] for text generation.

Datasets and evaluations. VTAB-1K comprises 19 datasets organized into (i) Natural images, (ii) Specialized datasets (remote sensing, medical) and (iii) Structured datasets (scene structure) domains. Each dataset has 1K training examples. Following [78, 28], we use the provided 800-200 train split for hyperparameter selection, evaluate using the full training set and report average accuracy across three trails. **Few-shot learning** involves 5 fine-grained datasets: FGVC-Aircraft [46], Food101 [4], OxfordFlowers102 [51], OxfordPets [53] and StanfordCars [35]. Following [30], we evaluate 1,

Table 1: Results on VTAB-1K with ViT-B/16. Mean Acc. is the average of group mean values.

Method	Natural						Specialized				Structured								Mean Acc.	
	Cifar100	Caltech101	DTD	Flowers102	Pets	SVHN	Sun397	Camelyon	EuroSAT	Resisc45	Retinopathy	Clevr-Count	Clevr-Dist	DMLab	KITTI-Dist	dSpr-Loc	dSpr-Ori	sNORB-Azrim		NsORB-Ele
Full	68.9	87.7	64.3	97.3	86.9	87.4	38.8	79.7	95.7	84.2	73.9	56.3	58.6	41.7	65.5	57.5	46.7	25.7	29.1	68.9
Linear	64.4	85.0	63.2	97.0	86.3	36.6	51.0	78.5	87.5	68.5	74.0	34.3	30.6	33.2	55.4	12.5	20.0	9.6	19.2	57.6
VPT-Deep	78.8	90.8	65.8	98.0	88.3	78.1	49.6	81.8	96.1	83.4	68.4	68.5	60.0	46.5	72.8	73.6	47.9	32.9	37.8	72.0
Adapter	69.2	90.1	68.0	98.8	89.9	82.8	54.3	84.0	94.9	81.9	75.5	80.9	65.3	48.6	78.3	74.8	48.5	29.9	41.6	73.9
AdaptFormer	70.8	91.2	70.5	99.1	90.9	86.6	54.8	83.0	95.8	84.4	76.3	81.9	64.3	49.3	80.3	76.3	45.7	31.7	41.1	74.7
LoRA	67.1	91.4	69.4	98.8	90.4	85.3	54.0	84.9	95.3	84.4	73.6	82.9	69.2	49.8	78.5	75.7	47.1	31.0	44.0	74.5
NOAH	69.6	92.7	70.2	99.1	90.4	86.1	53.7	84.4	95.4	83.9	75.8	82.8	68.9	49.9	81.7	81.8	48.3	32.8	44.2	74.2
RepAdapter	69.0	92.6	75.1	99.4	91.8	90.2	52.9	87.4	95.9	87.4	75.5	75.9	62.3	53.3	80.6	77.3	54.9	29.5	37.9	76.1
RLRR	75.6	92.4	72.9	99.3	91.5	89.8	57.0	86.8	95.2	85.3	75.9	79.7	64.2	53.9	82.1	83.9	53.7	33.4	43.6	76.7
GLoRA	76.4	92.9	74.6	99.6	92.5	91.5	57.8	87.3	96.8	88.0	76.0	83.1	67.3	54.5	86.2	83.8	52.9	37.0	41.4	78.0
Baseline	74.9	93.3	72.0	99.4	91.0	91.5	54.8	83.2	95.7	86.9	74.2	83.0	70.5	51.9	81.4	77.9	51.7	33.6	44.4	76.4
+PACE	79.0	94.2	73.6	99.4	92.4	93.7	58.0	87.4	96.4	89.3	77.1	84.9	70.9	54.9	84.3	84.7	57.3	39.3	44.8	79.0

Table 2: Classification accuracy on Few-shot learning with ViT-B/16 pre-trained on ImageNet-21K.

Method \ Shot	FGVCAircraft					Food101					Flowers102				
	1	2	4	8	16	1	2	4	8	16	1	2	4	8	16
LoRA _{add}	10.4	15.2	27.2	41.7	59.2	33.9	51.9	59.3	66.0	71.3	93.3	96.4	98.0	98.6	98.7
+PACE	10.7	16.3	28.2	42.1	61.0	40.6	55.9	63.8	70.3	75.2	95.0	98.0	98.9	99.5	99.6
VPT _{add}	11.2	15.1	23.7	36.3	51.5	34.3	56.6	64.8	71.7	75.4	94.3	97.6	98.2	99.3	99.6
+PACE	11.6	16.2	24.0	37.0	52.4	39.9	57.2	66.7	72.4	76.1	95.3	97.8	98.6	99.4	99.6
LoRA _{mul} +VPT _{add}	10.5	15.6	28.4	44.8	61.8	35.4	54.3	64.8	72.1	76.4	90.4	97.3	98.4	99.4	99.5
+PACE	12.3	16.8	29.9	45.7	62.5	39.3	57.2	66.7	73.4	77.8	93.4	98.1	99.1	99.5	99.7
OxfordPets					StanfordCars					Average					
LoRA _{add}	73.2	83.1	87.5	89.2	91.1	8.7	15.3	30.2	55.3	74.5	43.9	52.3	60.4	70.1	78.9
+PACE	75.3	85.0	90.7	90.8	92.4	9.4	16.0	30.9	56.1	75.9	46.2	54.2	62.5	71.7	80.8
VPT _{add}	75.9	85.6	90.3	90.6	92.3	9.3	15.0	27.8	46.6	65.1	45.0	53.9	60.9	68.9	76.7
+PACE	78.2	87.4	90.3	91.1	92.3	9.9	15.4	27.9	47.0	65.9	46.9	54.8	61.5	69.3	77.2
LoRA _{mul} +VPT _{add}	69.9	84.1	89.1	91.3	91.9	9.0	16.3	32.7	59.0	76.4	43.0	53.5	62.6	73.2	81.2
+PACE	76.5	88.0	90.3	91.4	92.4	9.7	16.4	33.7	59.8	77.3	46.2	55.3	63.9	73.9	81.9

2, 4, 8 and 16 shots, train on the provided training set, tune hyperparameters using validation and report average test accuracy over three random seeds. **FGVC** includes 5 fine-grained datasets: CUB-200-2011 [69], NABirds [66], OxfordFlowers [51], StanfordDogs [10] and StanfordCars [35]. We follow [28] to use validation set for hyperparameter and report test results. For **domain adaptation**, following [82, 7], we train on ImageNet [11] with a 16-shot setting, use the validation split by [82] for hyperparameter selection and report the results on the official validation set and 4 out-of-domain datasets: ImageNet-Sketch [71], ImageNet-V2 [56], ImageNet-A [23] and ImageNet-R [22]. We evaluate on GLUE [70] for **text classification** and GSM-8K [9] for **mathematical reasoning**.

Pre-trained backbones. We experiment with two vision transformers, Vision Transformers (ViT-B/16) [16] and Swin Transformer (Swin-B) [44]. These two are pre-trained on ImageNet-21K [11]. We test a ViT-B-Laion-IN12K model, pre-trained on Laion-2B [60] and fine-tuned on ImageNet-12K [11]. We use RoBERTa_{base} [43] and Phi-3-mini-4k-instruct [1] for text classification and generation.

Implementation details. We follow [28] for image processing: 224×224 resizing for VTAB-1K; random flips and crops to 224×224 for FGVC and few-shot learning; stronger augmentation for domain adaptation task, following [16, 82, 41]. We use the Adam optimizer [32] with cosine learning rate decay and linear warm-up (first 10 epochs). Models are fine-tuned for 300 epochs on VTAB-1K and 100 epochs on other vision adaptation tasks, with batch size 64. For text classification we follow [26]. See §G for mathematical reasoning details. All experiments used an NVIDIA H100 GPU.

Baseline. For each dataset, we identified the better method (LoRA_{mul}+VPT_{add} or LoRA_{add}) and tuned the rank, learning rate, and weight decay to form a strong baseline. The detailed baseline settings for each task and the number of trainable parameters are provided in §F, where LoRA_{mul}+VPT_{add} generally outperformed other variants. Building on the strong LoRA_{mul}+VPT_{add}, we use the grid search for our λ and σ , following strategies from previous studies [28, 41, 26]. Beyond LoRA_{mul}+VPT_{add}, PACE also enhances PEFT methods such as AdaptFormer, GLoRA, COFT, and BOFT (§D.4).

Table 3: Results on FGVC with ViT-B/16.

* denotes using augmented ViT by AugReg [62].

Method	CUB -2011	NA- Birds	Oxford Flowers	Stan. Dogs	Stan. Cars	Mean Acc.
Full	87.3	82.7	98.8	89.4	84.5	85.9
Linear	85.3	75.9	97.9	86.2	51.3	79.3
VPT	88.5	84.2	99.0	90.2	83.6	89.1
LoRA	88.3	85.6	99.2	91.0	83.2	89.5
SSF*	89.5	85.7	99.6	89.6	89.2	90.7
ARC*	89.3	85.7	99.7	89.1	89.5	90.7
RLRR*	89.8	85.3	99.6	90.0	90.4	91.0
LoRA _{mul} +VPT _{add}	88.9	87.1	99.4	91.2	87.5	90.8
+PACE	89.8	87.3	99.5	92.2	88.8	91.5

Table 4: Results on domain adaptation with ViT-B/16 pre-trained on ImageNet-21K.

Method	Source	Target				Mean Acc.
	ImageNet	-Sketch	-V2	-A	-R	
Full	63.9	18.5	52.5	3.2	21.2	31.8
Linear	67.9	14.4	60.8	9.4	25.6	35.6
Adapter	70.5	16.4	59.1	5.5	22.1	34.7
VPT	70.5	18.3	58.0	4.6	23.2	34.7
LoRA	70.8	20.0	59.3	6.9	23.3	36.0
NOAH	71.5	24.8	66.1	11.9	28.5	40.5
GLoRA	78.3	30.6	67.5	13.3	31.0	44.1
LoRA _{mul} +VPT _{add}	78.3	30.6	68.5	14.1	32.5	44.8
+PACE	79.0	31.8	69.4	16.3	35.2	46.3

Table 5: Results for GLUE w/ RoBERTa_{base}. Matthew’s correlation for COLA, Pearson correlation for STSB, and accuracy for others.

Method	COLA	STSB	MRPC	RTE	QNLI	SST2	Avg.
Full	63.6	91.2	90.2	78.7	92.8	94.8	85.2
BitFit	62.0	90.8	92.7	81.5	91.8	93.7	85.4
Adapt	62.6	90.3	88.4	75.9	93.0	94.7	84.2
VeRA	65.6	90.7	89.5	78.7	91.8	94.6	85.2
LoRA	63.4	91.5	89.7	86.6	93.3	95.1	86.6
+PACE	66.2	92.0	91.4	86.9	93.6	95.6	87.6

Table 6: Results for GSM-8K using Phi-3-mini-4k-instruct.

Method	Accuracy
Pre-trained	62.01
Full	73.16
LoRA	75.66
+PACE	78.77

Table 7: Classification results on domain adaptation and CIFAR-100 in VTAB-1K based different pre-trained models. Src. is short for ‘source’ in Table 4.

Method	ViT-B (ImageNet-21K)						ViT-B (Laion2B-ImageNet-12K)						Swin-B (ImageNet-21K)					
	CIFAR -100	Src.	-S	-V	-A	-R	CIFAR -100	Src.	-S	-V	-A	-R	CIFAR -100	Src.	-S	-V	-A	-R
Full	51.6	63.9	18.5	52.5	3.2	21.2	51.2	66.0	29.0	56.1	8.1	27.9	65.6	71.7	27.0	61.1	10.8	24.4
Linear	63.4	67.9	14.4	60.8	9.4	25.6	61.9	79.2	43.2	69.5	23.4	40.9	65.0	78.8	36.7	68.8	23.2	35.9
LoRA _{add}	71.2	73.8	27.1	64.8	13.6	25.0	71.3	77.5	39.8	67.8	20.4	35.6	74.3	76.3	30.7	65.7	16.8	28.9
VPT _{add}	73.6	74.3	27.1	65.9	11.5	26.7	71.8	78.4	40.4	68.7	22.4	38.4	72.7	76.2	30.6	66.2	17.6	29.1
LoRA _{mul}	73.4	78.1	31.2	68.3	13.4	32.7	73.2	78.6	41.9	68.8	22.6	37.8	73.9	76.1	30.8	65.7	18.1	28.9
LoRA _{add} +VPT _{add}	70.3	76.8	28.7	66.6	13.7	29.9	71.8	78.0	41.4	68.3	20.6	36.9	74.5	76.3	30.7	65.7	16.8	28.9
LoRA _{mul} +VPT _{add}	74.9	78.3	30.6	68.5	14.1	32.5	73.8	78.3	41.5	68.6	21.6	38.2	74.6	76.6	31.2	66.5	18.5	29.4
+PACE	79.0	79.0	31.8	69.4	16.3	35.2	78.0	80.1	45.8	71.2	24.6	43.6	78.9	79.6	39.2	70.1	25.2	38.0

4.1 Comparison with the State of the Arts

Results on VTAB-1K. Table 1 presents the results comparing PACE with recent state-of-the-art PEFT methods. PACE improves the strong baseline by 2.6% accuracy, surpassing the previous SOTA GLoRA [7] by 1%, which uses two stages for parameter search. In §D.1, we show that reducing training epochs to 50 or 100 has minimal impact on PACE performance.

Results on Few-shot Learning. Table 2 compares performance w/ and w/o our PACE. PACE improves LoRA_{add}, VPT_{add}, LoRA_{mul}+VPT_{add}, with LoRA_{mul}+VPT_{add}+PACE performing best in most cases. PACE yields notable improvement, especially when the number of shot is small.

Results on FGVC. Table 3 shows that PACE improves the strong LoRA_{mul}+VPT_{add} by 0.7%, outperforming SSF [41], ARC [14] and RLRR [15] that use strongly pre-trained ViT with augmentations. In §D.2, PACE achieves larger improvements on smaller datasets.

Results on domain adaptation. Table 4 compares PACE with others. LoRA_{mul}+VPT_{add} outperforms GLoRA [7] which relies on parameter search. Meanwhile, PACE improves LoRA_{mul}+VPT_{add} by 1.5%, outperforming other PEFT methods, demonstrating superior performance on domain adaptation.

Results on text classification and mathematical reasoning. Table 5 shows that PACE outperforms LoRA by 1% on GLUE text classification and by 3.11% on GSM-8K mathematical reasoning.

Generalization on other backbones. We evaluate PACE on CIFAR-100 (VTAB-1K) and domain adaptation using Swin-B [44] pre-trained on ImageNet-21K and ViT-B (pre-trained on Laion 2B, then fine-tuned on ImageNet-12K). Table 7 shows PACE outperforms LoRA_{mul}+VPT_{add} and other PEFT methods across all backbones, demonstrating its strong generalizability. Further experiments in §D.3 show PACE works effectively with self-supervised models such as MAE [19] and DINO [5].

4.2 Analyses

To verify our theories, we conduct experiments on CIFAR-100 (VTAB-1K) using ViT-B/16 and Camelyon (VTAB-1K) on Swin-B. Figures 3 & 4 show the gradient norm (summed across all layers) and FP-distance (Eq. 5) and the train & validation accuracy during training for baseline $\text{LoRA}_{\text{mul}}+\text{VPT}_{\text{add}}$ and PACE on validation set. Figures 3a & 4a show that PACE has a smaller gradient norm than baseline, verifying Theorem 2 that PACE can implicitly lower the weight gradient norm for better generalization. Figures 3b & 4b demonstrate that PACE maintains a lower FP-distance than the baseline, verifying Theorem 3 that PACE can implicitly align the fine-tuned model with pre-trained model, retaining knowledge from large-scale pre-training. Owing to the advantages of the gradient regularization and model alignment, PACE shortens the performance gap between seen and unseen data, yielding higher accuracy on the unseen validation set, as shown in Figures 3c & 4c.

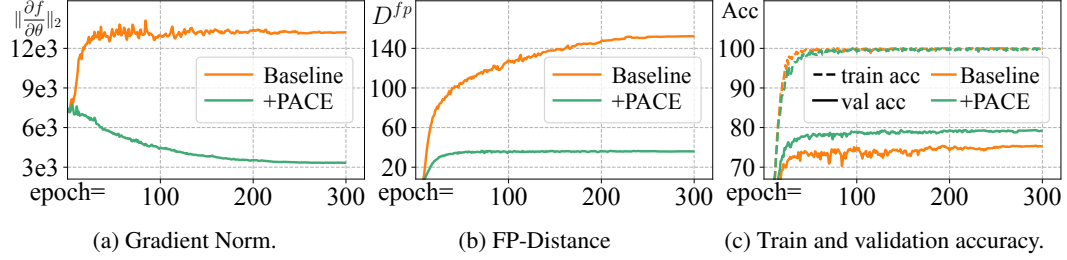


Figure 3: Analysis for PACE. (a) gradient norm, (b) FP-Distance and (c) train & val. accuracy are evaluated on validation set of CIFAR-100 (VTAB-1K) with baseline $\text{LoRA}_{\text{mul}}+\text{VPT}_{\text{add}}$ on ViT-B/16.

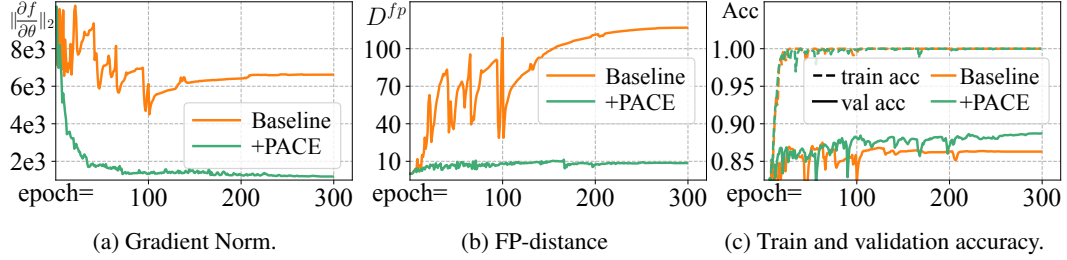


Figure 4: Analysis for PACE. (a) gradient norm, (b) FP-Distance and (c) train & val. accuracy are evaluated on the validation set of Camelyon (VTAB-1K) with baseline $\text{LoRA}_{\text{mul}}+\text{VPT}_{\text{add}}$ on Swin-B.

To clarify why naive alignment is problematic, we vary the regularization strength λ over a wide range ($1e-3$ to $5e4$) for both Fine-tuned Pre-trained model Alignment (FPA) by minimizing D^{fp} in Eq. 5 and PACE. Figure 5 shows the averaged gradient norm over training (see also Figures 8 & 9 for more visualizations). PACE robustly lowers gradient norms with larger λ , while FPA exhibits unpredictable behavior, even causing gradient explosion. This verifies Prop. 1 that minimizing D^{fp} is problematic for gradient regularization, complicating gradient management.

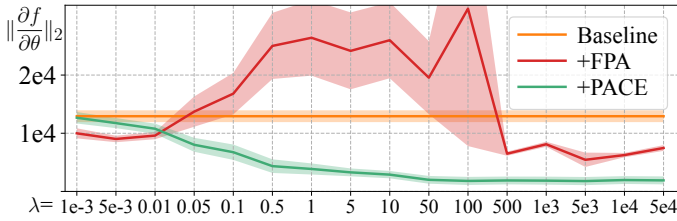


Figure 5: Gradient norms of models across wide range of regularization strengths λ on CIFAR-100 (VTAB-1K) w/ ViT-B/16. Line and shadow represent mean and std across training epochs.

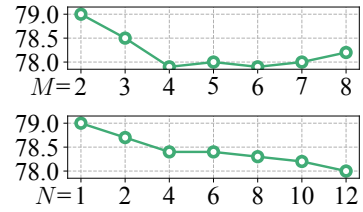


Figure 6: Ablation results for applying PACE among M nets and lazily at every N steps.

4.3 Ablation studies

We ablate PACE based on the baseline $\text{LoRA}_{\text{mul}}+\text{VPT}_{\text{add}}$ on CIFAR-100 (VTAB-1K) and ImageNet-1K in domain adaption as shown in Table 8. The ablations include Noise (baseline w/ noise perturbing

adapter), PACE_{add} (replacing the multiplicative noise with the additive noise), PACE_h (perturbing $h(\cdot)$ instead of $\Delta h(\cdot)$ in Eq. 11), $\text{PACE}_{\text{drop}}$ (replacing the Gaussian noise with the dropout noise), $\text{PACE}_{\sigma=}$ (all transformer blocks share the same σ), $\text{PACE}_{\sigma\uparrow}$ (σ increases linearly with depth), FPA (fine-tuned and pre-trained alignment by minimizing Eq. 5), SAM (sharpness-aware minimization [17]), GP (gradient penalization), ℓ_1 (sparsity regularization), and transfer learning methods L2SP [77], DELTA [40] and FTP [64]. We grid-search hyperparameters and report the best results.

Table 8 presents the results for all variants. PACE improves over Noise, which itself is better than baseline, justifying our adapter perturbation and consistency regularization. PACE_{add} performs worse than PACE, showing the superiority of the multiplicative noise. Although PACE_h can implicitly regularize gradients, it performs worse than PACE, verifying the advantages of perturbing adapter to implicitly align models. $\text{PACE}_{\text{drop}}$ is worse than PACE, indicating the dropout noise is suboptimal. $\text{PACE}_{\sigma=}$ and $\text{PACE}_{\sigma\uparrow}$ perform worse, justifying our design of linearly decreasing σ . FPA, SAM and GP, which either only align models or only regularize gradients, are outperformed by PACE. Despite combining FPA+GP, it still performs worse than ours, suggesting ineffective combination. ℓ_1 , L2SP, DELTA, and FTP obtain worse results than PACE, showing their limitations in improving generalization. PACE regularizes gradients for better generalization and aligns models to retain knowledge, surpassing all other variants.

Method	CIFAR -100	ImageNet-1K				
		Source	-Sketch	-V2	-A	-R
LoRA _{mul} +VPT _{add}	74.9	78.3	30.6	68.5	14.1	32.5
+Noise	77.4	78.3	31.3	68.6	14.3	33.0
+PACE	79.0	79.0	31.8	69.4	16.3	35.2
+ PACE_{add}	75.7	78.3	31.2	68.7	13.7	32.7
+ PACE_h	75.9	78.4	31.2	68.1	13.8	32.6
+ $\text{PACE}_{\text{drop}}$	78.3	78.9	31.2	68.9	16.0	34.6
+ $\text{PACE}_{\sigma=}$	77.9	78.8	31.6	68.3	16.6	34.7
+ $\text{PACE}_{\sigma\uparrow}$	77.3	78.7	31.3	68.9	14.0	33.6
+FPA	76.6	78.8	31.2	68.6	14.7	33.5
+SAM [17]	75.4	78.4	31.4	68.5	13.8	32.9
+GP	75.8	78.3	31.7	68.4	14.2	32.1
+FPA+GP	74.9	78.1	31.5	68.1	13.5	32.6
+ ℓ_1	75.2	78.2	30.6	68.6	13.7	32.8
+L2SP [77]	75.9	78.5	30.4	68.7	14.9	33.5
+DELTA [40]	76.4	78.4	30.8	68.7	14.6	33.7
+FTP [64]	76.2	78.6	30.8	68.6	15.8	33.6

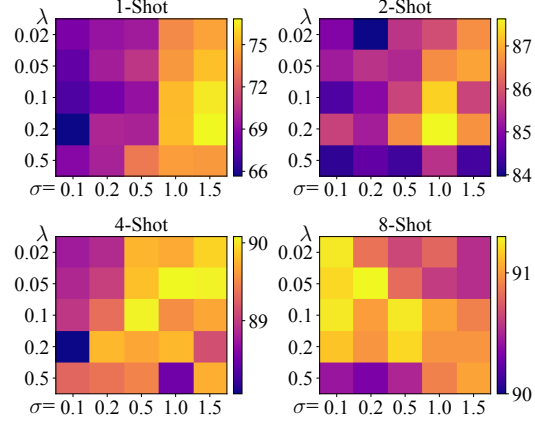


Table 8: Accuracy results on domain adaptation and VTAB-1K based different pre-trained models. Figure 7: Results for varied λ and σ as well as shot on OxfordPets in few-shot learning.

We further evaluate applying PACE across multiple M networks during training or applying it lazily with half-batch size at every N steps ($\text{PACE}_{\text{lazy}}^{\text{half}}$ in §C). Figure 6 presents the results, showing that applying PACE among two networks at every training step performs best. However, lazy regularization applied every few steps can still provide reasonable results while saving computational/memory costs.

We test the sensitivity of hyperparameters λ and σ introduced in our PACE on OxfordPets for few-shot learning across 1, 2, 4, 8 shots. The results presented in Figure 7 demonstrate that with less data, larger λ and σ are favored, verifying the effectiveness of PACE in improving generalization.

5 Conclusions

We have introduced PACE, a novel and effective method that combines generalization of Parameter-efficient fine-tuning with Consistency rEgularization. Through rigorous theoretical analyses, we have shown PACE reduces weight gradient for improved generalization and it aligns the fine-tuned model with the pre-trained model for retaining pre-training knowledge. Our experimental results support the theoretical analyses, justifying the generalization advantages of PACE over other PEFT methods. With its dual advantages, PACE consistently outperforms other variants across different backbones, firmly establishing PACE as a powerful solution for enhancing generalization for PEFT methods. Limitations and border impacts are discussed in §A.

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PACE: Marrying generalization of PArAmeter-efficient fine-tuning with Consistency rEgularization (Supplementary Material)

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A Broader impacts and limitations

A.1 Broader impacts

Our work provides a powerful solution for improving generalization in Parameter Efficient Fine-Tuning (PEFT), allowing for effective fine-tuning of pre-trained models while reducing the heavily reliance on pre-training from scratch using large scale data. Our advancements in PEFT, supported by Theorems 1, 2 and 3, offer novel insights into gradient regularization and model alignment. These insights extend beyond PEFT and can be applied to other areas such as continual learning and transfer learning, potentially enhancing the performance and efficiency of models in various domains. By leveraging our findings, practitioners can develop more robust and adaptable models that generalize well to new tasks and environments, leading to more intelligent and versatile AI systems. In terms of negative impacts, the robustness of our fine-tuning method could potentially be misused to create more convincing deepfakes, raising concerns about the spread of misinformation, manipulation of public opinion, and malicious activities such as fraud, blackmail, or harassment. However, potential misuse is a downside with any improvements that have universal nature.

A.2 Limitations

While our work effectively improves generalization ability, it introduces additional computational costs by requiring input samples to be passed through the network twice for regularization. However, this can be mitigated by using two efficient variants, PACE_{fast} and PACE_{lazy}^{half}, proposed in §C, where we demonstrate the potential for resource-efficient fine-tuning. Additionally, our method introduces extra hyperparameters λ and σ , which require caution during hyperparameter search. Nonetheless, Figure 7 suggests that fewer training data requires larger λ and σ values, providing insight for hyperparameter tuning.

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B Proofs

B.1 Proof of Theorem 1

Setting $\epsilon = \frac{\rho \nabla_{\theta}}{\|\nabla_{\theta}\|_2}$, we perform a second-order Taylor expansion of $\mathcal{L}_{\mathcal{D}^n}$ around θ . By incorporating the higher-order terms from the Taylor expansion into $R\left(\frac{\|\theta\|_2^2}{\rho^2}, \frac{1}{n}\right)$, we derive:

$$\begin{aligned}\mathcal{L}_{\mathcal{D}}(\theta) &\leq \mathcal{L}_{\mathcal{D}^n}\left(\theta + \frac{\rho \nabla_{\theta}}{\|\nabla_{\theta}\|_2}\right) + R\left(\frac{\|\theta\|_2^2}{\rho^2}, \frac{1}{n}\right) \\ &\approx \mathcal{L}_{\mathcal{D}^n}(\theta) + \rho \|\nabla_{\theta}\|_2 + \frac{\rho^2}{2\|\nabla_{\theta}\|_2^2} \nabla_{\theta}^T \mathbf{H}_{\theta} \nabla_{\theta} + R\left(\frac{\|\theta\|_2^2}{\rho^2}, \frac{1}{n}\right).\end{aligned}\quad (14)$$

Assuming that the approximation does not alter the inequality relationship, *i.e.*, it preserves the $\leq$ relation on both sides and considering the largest eigenvalue of $\mathbf{H}_{\theta}$ as $\lambda_{\max}^{\mathbf{H}}$, implying $\mathbf{v}^T \mathbf{H}_{\theta} \mathbf{v} \leq \lambda_{\max}^{\mathbf{H}} \|\mathbf{v}\|_2^2$ for any $\mathbf{v}$, we further bound Eq. 14 as follows and arrive at:

$$\mathcal{L}_{\mathcal{D}}(\theta) \leq \mathcal{L}_{\mathcal{D}^n}(\theta) + \rho \|\nabla_{\theta}\|_2 + \frac{\rho^2}{2} \lambda_{\max}^{\mathbf{H}} + R\left(\frac{\|\theta\|_2^2}{\rho^2}, \frac{1}{n}\right).$$

B.2 Proof of Theorem 2

The proof is motivated by Ni and Koniusz [47]. We include the proof process for completeness. Denote $\mathbf{m}_1 = \mathbf{z}_1 - \mathbf{1}$, $\mathbf{m}_2 = \mathbf{z}_2 - \mathbf{1}$ thus $\mathbf{m}_1, \mathbf{m}_2 \sim \mathcal{N}(\mathbf{0}, \sigma^2)$

$$\begin{aligned}d^{\text{pace}} &= \mathbb{E}_{\mathbf{z}_1, \mathbf{z}_2} [f(\theta_0 + \mathbf{z}_1 \odot \Delta\theta) - f(\theta_0 + \mathbf{z}_2 \odot \Delta\theta)]^2 \\ &= \mathbb{E}_{\mathbf{z}_1, \mathbf{z}_2} [f(\theta_0 + \Delta\theta + (\mathbf{z}_1 - \mathbf{1}) \odot \Delta\theta) - f(\theta_0 + \Delta\theta + (\mathbf{z}_2 - \mathbf{1}) \odot \Delta\theta)]^2 \\ &= \mathbb{E}_{\mathbf{m}_1, \mathbf{m}_2} [f(\theta + \mathbf{m}_1 \odot \Delta\theta) - f(\theta + \mathbf{m}_2 \odot \Delta\theta)]^2.\end{aligned}\quad (15)$$

Defining $\mathbf{v} := \mathbf{m}_1 \odot \Delta\theta$ and $\mathbf{u} := \mathbf{m}_2 \odot \Delta\theta$, where $\mathbf{v}, \mathbf{u} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \text{diag}(\Delta\theta \odot \Delta\theta))$, we can rewrite Eq. 15 as follows:

$$\begin{aligned}&\mathbb{E}_{\mathbf{v}, \mathbf{u}} [f(\theta + \mathbf{v}) - f(\theta + \mathbf{u})]^2 \\ &\approx \mathbb{E}_{\mathbf{v}, \mathbf{u}} \left[f(\theta) + \mathbf{v}^T \nabla + \frac{1}{2} \mathbf{v}^T \mathbf{H} \mathbf{v} - f(\theta) - \mathbf{u}^T \nabla - \frac{1}{2} \mathbf{u}^T \mathbf{H} \mathbf{u} \right]^2 \\ &= \mathbb{E}_{\mathbf{v}, \mathbf{u}} \left[\mathbf{v}^T \nabla + \frac{1}{2} \mathbf{v}^T \mathbf{H} \mathbf{v} - \mathbf{u}^T \nabla - \frac{1}{2} \mathbf{u}^T \mathbf{H} \mathbf{u} \right]^2 \\ &= \mathbb{E}_{\mathbf{v}, \mathbf{u}} \left[(\mathbf{v} - \mathbf{u})^T \nabla + \frac{1}{2} \mathbf{v}^T \mathbf{H} \mathbf{v} - \frac{1}{2} \mathbf{u}^T \mathbf{H} \mathbf{u} \right]^2 \\ &= \mathbb{E}_{\mathbf{v}, \mathbf{u}} [(\mathbf{v} - \mathbf{u})^T \nabla]^2\end{aligned}\quad (16)$$

$$+ \mathbb{E}_{\mathbf{v}, \mathbf{u}} [((\mathbf{v} - \mathbf{u})^T \nabla)(\mathbf{v}^T \mathbf{H} \mathbf{v} - \mathbf{u}^T \mathbf{H} \mathbf{u})] \quad (17)$$

$$+ \frac{1}{4} \mathbb{E}_{\mathbf{v}} [\mathbf{v}^T \mathbf{H} \mathbf{v}]^2 + \frac{1}{4} \mathbb{E}_{\mathbf{u}} [\mathbf{u}^T \mathbf{H} \mathbf{u}]^2 \quad (18)$$

$$- \frac{1}{2} \mathbb{E}_{\mathbf{v}, \mathbf{u}} [(\mathbf{v}^T \mathbf{H} \mathbf{v})(\mathbf{u}^T \mathbf{H} \mathbf{u})]. \quad (19)$$

Next, we derive the four terms, Eq. 16, 17, 18, and 19, respectively as follows:

Eq. 16. Using $\mathbb{E}_{\mathbf{z}_1, \mathbf{z}_2} [(z_1 - z_2)^2] = 2\sigma^2$ for $\mathbf{z}_1, \mathbf{z}_2 \sim \mathcal{N}(\mathbf{0}, \sigma^2)$, we can simplify (Eq. 16) as follows, noting that terms related to different dimensions are canceled due to zero-mean independent Gaussian noise:

$$\mathbb{E}_{\mathbf{v}, \mathbf{u}} [(\mathbf{v} - \mathbf{u})^T \nabla]^2 = \mathbb{E}_{\mathbf{v}, \mathbf{u}} \left[\sum_j (v_j - u_j)^2 \nabla_j^2 \right] = 2\sigma^2 \sum_j \Delta\theta_j^2 \nabla_k^2. \quad (20)$$

Eq. 17. Utilizing $E[z^3] = \mu^3 + 3\mu\sigma^2$ for $z \sim \mathcal{N}(\mu, \sigma^2)$, and noting that $E[z^3] = 0$ for $\mu = 0$, Eq. 17 is derived as:

$$\begin{aligned}&\mathbb{E}_{\mathbf{v}, \mathbf{u}} [((\mathbf{v} - \mathbf{u})^T \nabla)(\mathbf{v}^T \mathbf{H} \mathbf{v} - \mathbf{u}^T \mathbf{H} \mathbf{u})] \\ &= \mathbb{E}_{\mathbf{v}} [(\mathbf{v}^T \nabla)(\mathbf{v}^T \mathbf{H} \mathbf{v})] + \mathbb{E}_{\mathbf{u}} [(\mathbf{u}^T \nabla)(\mathbf{u}^T \mathbf{H} \mathbf{u})] - \mathbb{E}_{\mathbf{v}, \mathbf{u}} [(\mathbf{v}^T \nabla)(\mathbf{u}^T \mathbf{H} \mathbf{u})] - \mathbb{E}_{\mathbf{v}, \mathbf{u}} [(\mathbf{u}^T \nabla)(\mathbf{v}^T \mathbf{H} \mathbf{v})] \\ &= 2\mathbb{E}_{\mathbf{v}} [(\mathbf{v}^T \nabla)(\mathbf{v}^T \mathbf{H} \mathbf{v})] = 0.\end{aligned}\quad (21)$$

Eq. 18. We first decompose Eq. 18, then discuss each case and obtain the final result:

$$\frac{1}{4}\mathbb{E}_{\mathbf{v}}[\mathbf{v}^T \mathbf{H} \mathbf{v}]^2 + \frac{1}{4}\mathbb{E}_{\mathbf{u}}[\mathbf{u}^T \mathbf{H} \mathbf{u}]^2 = \frac{1}{2}\mathbb{E}_{\mathbf{v}}[\mathbf{v}^T \mathbf{H} \mathbf{v}]^2 = \frac{1}{2}\mathbb{E}_{\mathbf{v}}\left[\sum_{j,k,p,q} v_j H_{jk} v_k v_p H_{pq} v_q\right]. \quad (22)$$

Given the independence of elements in $\mathbf{v}$, only terms with an element repeated two or four times contribute non-zero results, leading to four distinct, non-overlapping cases. Using $\mathbb{E}[z^2] = \sigma^2 + \mu^2$ and $\mathbb{E}[z^4] = \mu^4 + 6\mu^2\sigma^2 + 3\sigma^4$ for $z \sim \mathcal{N}(\mu, \sigma^2)$, and simplifying to $\mathbb{E}[z^2] = \sigma^2$ and $\mathbb{E}[z^4] = 3\sigma^4$ when $\mu = 0$, we have:

Case 1: $j = k \neq p = q$, given the independence of v_j and v_p , we have:

$$\mathbb{E}_{\mathbf{v}}\left[\sum_j \sum_{p \neq j} v_j^2 H_{jj} v_p^2 H_{pp}\right] = \sum_{j,p \neq j} H_{jj} H_{pp} \mathbb{E}[v_j^2] \mathbb{E}[v_p^2] = \sigma^4 \sum_{j,k \neq j} H_{jj} H_{kk} \Delta\theta_j^2 \Delta\theta_k^2. \quad (23)$$

Case 2: For $j = p \neq k = q$, the independence of v_j and v_k simplifies our calculation, leading to:

$$\mathbb{E}_{\mathbf{v}}\left[\sum_j \sum_{k \neq j} v_j H_{jk} v_k v_j H_{jk} v_k\right] = \sum_{j,k \neq j} H_{jk}^2 \mathbb{E}[v_j^2] \mathbb{E}[v_k^2] = \sigma^4 \sum_{j,k \neq j} H_{jk}^2 \Delta\theta_j^2 \Delta\theta_k^2. \quad (24)$$

Case 3: For $j = q \neq k = p$, utilizing the independence of v_j and v_k as well as the symmetry $H_{jk} = H_{kj}$, we obtain:

$$\mathbb{E}_{\mathbf{v}}\left[\sum_j \sum_{k \neq j} v_j H_{jk} v_k v_k H_{kj} v_j\right] = \sum_{j,k \neq j} H_{jk}^2 \mathbb{E}[v_j^2] \mathbb{E}[v_k^2] = \sigma^4 \sum_{j,k \neq j} H_{jk}^2 \Delta\theta_j^2 \Delta\theta_k^2. \quad (25)$$

Case 4: For $j = q = k = p$, using $\mathbb{E}[z^4] = 3\sigma^4$ where $z \sim \mathcal{N}(0, \sigma^2)$, we have:

$$\mathbb{E}_{\mathbf{v}}\left[\sum_j v_j H_{jj} v_j v_j H_{jj} v_j\right] = \sum_j H_{jj}^2 \mathbb{E}[v_j^4] = 3\sigma^4 \sum_j H_{jj}^2 \Delta\theta_j^4. \quad (26)$$

Combining above four cases together, we have the result for Eq. 18:

$$\frac{\sigma^4}{2} \left(\sum_j 3H_{jj}^2 \Delta\theta_j^4 + \sum_{j,k \neq j} (H_{jj} H_{kk} + 2H_{jk}^2) \Delta\theta_j^2 \Delta\theta_k^2 \right). \quad (27)$$

Eq. 19:

$$\begin{aligned} & -\frac{1}{2}\mathbb{E}_{\mathbf{v},\mathbf{u}}[(\mathbf{v}^T \mathbf{H} \mathbf{v})(\mathbf{u}^T \mathbf{H} \mathbf{u})] \\ &= -\frac{1}{2}\mathbb{E}_{\mathbf{v}}[(\mathbf{v}^T \mathbf{H} \mathbf{v})] \mathbb{E}_{\mathbf{u}}[(\mathbf{u}^T \mathbf{H} \mathbf{u})] \\ &= -\frac{1}{2}\mathbb{E}_{\mathbf{v}}\left[\sum_j H_{jj} v_j^2\right] \mathbb{E}_{\mathbf{u}}\left[\sum_k H_{kk} v_k^2\right] \\ &= -\frac{1}{2}\left(\sum_j H_{jj} \mathbb{E}[v_j^2]\right) \left(\sum_k H_{kk} \mathbb{E}[v_k^2]\right) \\ &= -\frac{\sigma^4}{2}\left(\sum_j H_{jj}^2 \Delta\theta_j^4 + \sum_{j,k \neq j} H_{jj} H_{kk} \Delta\theta_j^2 \Delta\theta_k^2\right). \end{aligned} \quad (28)$$

With results of Eq. 20, 21, 27, 28, we have the final results:

$$\begin{aligned} d^{\text{pace}} &\approx 2\sigma^2 \sum_j \Delta\theta_j^2 \nabla_j^2 + 0 \\ &+ \frac{\sigma^4}{2} \left(\sum_j 3H_{jj}^2 \Delta\theta_j^4 + \sum_{j,k \neq j} (H_{jj} H_{kk} + 2H_{jk}^2) \Delta\theta_j^2 \Delta\theta_k^2 - \sum_j H_{jj}^2 \Delta\theta_j^4 - \sum_{j,k \neq j} H_{jj} H_{kk} \Delta\theta_j^2 \Delta\theta_k^2 \right) \\ &= 2\sigma^2 \sum_j \Delta\theta_j^2 \nabla_j^2 + \sigma^4 \left(\sum_j H_{jj}^2 \Delta\theta_j^4 + \sum_{j,k \neq j} H_{jk}^2 \Delta\theta_j^2 \Delta\theta_k^2 \right) \\ &= 2\sigma^2 \sum_j \Delta\theta_j^2 \nabla_j^2 + \sigma^4 \sum_{j,k} H_{jk}^2 \Delta\theta_j^2 \Delta\theta_k^2 = 2\sigma^2 \|\Delta\theta \odot \nabla\|_2^2 + \sigma^4 \|(\Delta\theta \Delta\theta^T) \odot \mathbf{H}\|_F^2. \end{aligned} \quad (29)$$

B.3 Proof of Theorem 3

The Cauchy-Schwarz inequality states that for $\mathbf{u}, \mathbf{v} \in \mathbb{R}^d$, we have $(\sum_j u_j v_j)^2 \leq (\sum_j u_j^2)(\sum_j v_j^2)$. Let $\mathbf{u} = \mathbf{1}$, it follows that $(\sum_j v_j)^2 \leq d \|\mathbf{v}\|_2^2$. Using this inequality, we then prove the following:

$$[\Delta \boldsymbol{\theta}^T \boldsymbol{\nabla} - \frac{1}{2} \Delta \boldsymbol{\theta}^T \mathbf{H} \Delta \boldsymbol{\theta}]^2 \leq 2[\Delta \boldsymbol{\theta}^T \boldsymbol{\nabla}]^2 + [\Delta \boldsymbol{\theta}^T \mathbf{H} \Delta \boldsymbol{\theta}]^2$$

$$[\Delta \boldsymbol{\theta}^T \boldsymbol{\nabla}]^2 = \left(\sum_j \Delta \theta_j \nabla_j \right)^2 \leq d \|\Delta \boldsymbol{\theta} \odot \boldsymbol{\nabla}\|_2^2. \quad (30)$$

$$[\Delta \boldsymbol{\theta}^T \mathbf{H} \Delta \boldsymbol{\theta}]^2 = \left(\sum_{j,k} \Delta \theta_j \Delta \theta_k H_{jk} \right)^2 \leq d^2 \|(\Delta \boldsymbol{\theta} \Delta \boldsymbol{\theta}^T) \odot \mathbf{H}\|_F^2 \quad (31)$$

Here, the inequality is obtained by treating $\Delta \theta_j \Delta \theta_k H_{jk}$ as an element of a vector with size of d^2 . This leads to the final results.

B.4 Rationale for one-dimensional output analysis

We use the squared L_2 distance for multi-dimensional outputs for D^{fp} and D^{pace} , which allows our one-dimensional analysis to naturally generalize to multiple dimensions. For example, for a vector-valued function in the naive alignment, $f(\boldsymbol{\theta}) = [f_1(\boldsymbol{\theta}), \dots, f_m(\boldsymbol{\theta})]$, where m is the output dimension, we have:

$$\|f(\boldsymbol{\theta}_0) - f(\boldsymbol{\theta}_0 + \Delta \boldsymbol{\theta})\|_2^2 = \sum_{i=1}^m [f_i(\boldsymbol{\theta}_0) - f_i(\boldsymbol{\theta}_0 + \Delta \boldsymbol{\theta})]^2.$$

This equality shows that the squared L_2 distance in multiple dimensions is simply the sum of non-negative squared differences in each dimension. Consequently, this additive nature enables our one-dimensional analysis to extend seamlessly to multiple dimensions in practice, aligning with our empirical observations.

B.5 R increases with $\frac{1}{n}$

According to [17], the function $R(\frac{\|\boldsymbol{\theta}\|_2^2}{\rho^2}, \frac{1}{n})$ in Eq. 3 is defined as:

$$R\left(\frac{\|\boldsymbol{\theta}\|_2^2}{\rho^2}, \frac{1}{n}\right) = \sqrt{\frac{k \log \left(1 + \frac{\|\boldsymbol{\theta}\|_2^2}{\rho^2} \left(1 + \sqrt{\frac{\log n}{k}}\right)^2\right) + 4 \log \frac{n}{\delta} + 8 \log(6n + 3k)}{n - 1}}.$$

Here k is the number of parameters, n is the number of training samples, $\delta \in (0, 1]$ is the confidence level and ρ is the max norm of the Gaussian perturbation noise.

To ensure R is valid, we require $n > 1$. To analyze how R changes with n , we fix $\frac{\|\boldsymbol{\theta}\|_2^2}{\rho^2}$ and break the expression under the square root of R into three terms:

$$R_1 = \frac{k \log \left(1 + \frac{\|\boldsymbol{\theta}\|_2^2}{\rho^2} \left(1 + \sqrt{\frac{\log n}{k}}\right)^2\right)}{n - 1}, \quad R_2 = \frac{4 \log n - 4 \log \delta}{n - 1}, \quad R_3 = \frac{8 \log(6n + 3k)}{n - 1}$$

We analyze each term separately to determine whether it decreases with increasing n .

Analysis for R_1 : The derivative for R_1 w.r.t. n is:

$$\begin{aligned}
R'_1 &= \frac{\frac{k}{1 + \frac{\|\theta\|_2^2}{\rho^2} \left(1 + \sqrt{\frac{\log n}{k}}\right)^2} \cdot 2 \frac{\|\theta\|_2^2}{\rho^2} \left(1 + \sqrt{\frac{\log n}{k}}\right) \cdot \frac{1}{2\sqrt{\frac{\log n}{k}}} \cdot \frac{1}{kn} \cdot (n-1) - k \log \left(1 + \frac{\|\theta\|_2^2}{\rho^2} \left(1 + \sqrt{\frac{\log n}{k}}\right)^2\right)}{(n-1)^2} \\
&= \frac{\frac{\frac{\|\theta\|_2^2}{\rho^2} \left(1 + \sqrt{\frac{\log n}{k}}\right)}{1 + \frac{\|\theta\|_2^2}{\rho^2} \left(1 + \sqrt{\frac{\log n}{k}}\right)^2} \cdot \frac{1}{\sqrt{\frac{\log n}{k}}} \cdot \frac{n-1}{n} - k \log \left(1 + \frac{\|\theta\|_2^2}{\rho^2} \left(1 + \sqrt{\frac{\log n}{k}}\right)^2\right)}{(n-1)^2} \\
&< \frac{\frac{\frac{\|\theta\|_2^2}{\rho^2} \left(1 + \sqrt{\frac{\log n}{k}}\right)}{\frac{\|\theta\|_2^2}{\rho^2} \left(1 + \sqrt{\frac{\log n}{k}}\right)^2} \cdot \frac{1}{\sqrt{\frac{\log n}{k}}} - k \log \left(\frac{\|\theta\|_2^2}{\rho^2} \left(1 + \sqrt{\frac{\log n}{k}}\right)^2\right)}{(n-1)^2} \\
&< \frac{\frac{1}{1 + \sqrt{\frac{\log n}{k}}} \cdot \frac{1}{\sqrt{\frac{\log n}{k}}} - k \left(\log \frac{\|\theta\|_2^2}{\rho^2} + \log \left(1 + \sqrt{\frac{\log n}{k}}\right)^2 \right)}{(n-1)^2} \\
&< \frac{\frac{1}{\sqrt{\frac{\log n}{k}}} \cdot \frac{1}{\sqrt{\frac{\log n}{k}}} - k \log \frac{\|\theta\|_2^2}{\rho^2} - k \log \left(1 + \sqrt{\frac{\log n}{k}}\right)^2}{(n-1)^2} \\
&= \frac{k}{(n-1)^2} \cdot \left(\frac{1}{\log n} - \log \frac{\|\theta\|_2^2}{\rho^2} - \log \left(1 + \sqrt{\frac{\log n}{k}}\right)^2 \right).
\end{aligned}$$

Since $\frac{\|\theta\|_2^2}{\rho^2}$ is generally large, the smallest n is 2 and $\log \left(1 + \sqrt{\frac{\log n}{k}}\right)^2 > 0$. Therefore, for $n > 1$, $R'_1 < 0$, meaning R_1 decreases as n increases.

Analysis of R_2 : The derivative for R_2 w.r.t. n is

$$R'_2 = \frac{4}{(n-1)^2} \left(1 - \frac{1}{n} - \log n + \log \delta\right).$$

Since $\delta \leq 1$, for $n > 1$, $R'_2 < 0$, indicating that R_2 decreases with increasing n .

Analysis of R_3 : The derivative for R_3 w.r.t. n is

$$R'_3 = \frac{8 \left(\frac{6(n-1)}{6n+3k} - \log(6n+3k) \right)}{(n-1)^2} < \frac{8(1 - \log(6n+3k))}{(n-1)^2}.$$

For $n > 1$, $\log(6n+3k) > 1$, implying that $R'_3 < 0$ and R_3 decrease as n increases.

Conclusion. For $n > 1$, all terms R_1 , R_2 and R_3 decreases as n increases. Thus $R(\frac{\|\theta\|_2^2}{\rho^2}, \frac{1}{n})$ is a decreasing function of n .

C Efficient PACE variants

Building upon strong theoretical foundation of PACE for generalization, we demonstrate that simple modifications can reduce memory and training time requirements of PACE. In this section, we explore two efficient variants, $\text{PACE}_{\text{fast}}$ and $\text{PACE}_{\text{lazy}}^{\text{half}}$, both maintaining similar computational and memory requirements as the baseline while improving performance. We then provide empirical results which show that $\text{PACE}_{\text{fast}}$ slightly outperforms $\text{PACE}_{\text{lazy}}^{\text{half}}$ while requiring no additional hyperparameters and using fewer computational resources. Given its superior efficiency, we further explore the potential of $\text{PACE}_{\text{fast}}$ for resource-efficient fine-tuning. By simply reducing the batch size and epochs, $\text{PACE}_{\text{fast}}$ outperforms the baseline while using significantly less GPU memory and training time.

$\text{PACE}_{\text{fast}}$: Building on the observation that only small datasets are typically available for fine-tuning, we assume that the model behavior changes gradually across epochs. Under this assumption, we store the model outputs from the previous epoch ($f_{e-1}(\mathbf{x})$), which contain inherent noise due to the adapter perturbation, and compute the consistency regularization loss between these stored outputs and the current epoch's noised outputs:

$$d_{\text{fast}}^{\text{pace}}(\mathbf{x}) = \|f(\mathbf{x}) - \mathbf{o}_{e-1}\|_2^2; \quad \text{where } \mathbf{o}_{e-1} = f_{e-1}(\mathbf{x}). \quad (32)$$

Here the output vector $\mathbf{o} \in \mathbb{R}^C$, where C is the number of classes. Since $f(\cdot)$ applies noise perturbation to the adapter and changes gradually between epochs, $f_{e-1}(\mathbf{x})$ and $f(\mathbf{x})$ can be seen as applying different *i.i.d.* noises to similar model states. This approach preserves the theoretical foundation of PACE while incurring minimal storage and computation costs. With typically few classes C and a limited number of samples in fine-tuning, storing $\mathbf{o}_{e-1}$ within GPU or CPU memory is manageable.

PACE_{lazy}^{half}. During training, the network always applies noise perturbations. Every N -th iteration uses a half batch size and consistency regularization, while all other iterations use the full batch size.

Memory and computational efficiency of two variants. Both variants maintain similar computational and memory requirements as the baseline. To demonstrate this, we conduct experiments on CIFAR-100 (VTAB-1K) using ViT-B/16, Camelyon (VTAB-1K) with Swin-B, and ImageNet (domain adaptation) with ViT-B/16. Table 9 compares maximum GPU memory usage, total training time, and accuracy for each task, showing that PACE_{fast} and PACE_{lazy}^{half} significantly improve upon the baseline while maintaining similar computational demands.

We find that PACE_{fast} slightly outperforms PACE_{lazy}^{half} without requiring additional hyperparameters, yet it needs to store outputs from the previous epoch. We therefore analyze its memory requirements.

Table 9: GPU memory usage, training time, and accuracy for PACE_{fast} and PACE_{lazy}^{half}. here, ‘m’ denotes minutes, Both variants outperform the baseline while maintaining similar computational demands.

Method	CIFAR-100 (ViT/16-B)			Camelyon (Swin-B)			ImageNet (ViT/16-B)		
	GPU Memory	Time	Accuracy	GPU Memory	Time	Accuracy	GPU Memory	Time	Mean Acc.
LoRA _{mul} +VPT _{add}	8.9GB	29m	74.6	15.7GB	33m	86.7	8.9GB	161m	44.8
+PACE	17.7GB	53m	79.0	29.4GB	60m	89.3	17.7GB	278m	46.3
+PACE _{fast}	9.0GB	29m	78.3	15.7GB	34m	88.8	9.0GB	162m	46.1
+PACE _{lazy} ^{half} ($N=2$)	9.3GB	29m	78.7	15.7GB	36m	89.2	9.0GB	165m	46.0
+PACE _{lazy} ^{half} ($N=4$)	9.3GB	29m	78.4	15.7GB	35m	88.9	9.0GB	163m	45.6
+PACE _{lazy} ^{half} ($N=6$)	9.3GB	29m	78.4	15.7GB	35m	89.0	9.0GB	163m	45.7
+PACE _{lazy} ^{half} ($N=10$)	9.3GB	29m	78.2	15.7GB	35m	88.9	9.0GB	162m	45.6

Memory efficiency of PACE_{fast}. We compare the additional memory requirement of PACE_{fast} with the baseline GPU memory consumption. Table 10 shows that the memory overhead of PACE_{fast} is negligible compared to the baseline GPU memory requirements and can be easily stored in GPU. Moreover, even in the rare scenario of fine-tuning on the full ImageNet 1K dataset (1.2 million samples), PACE_{fast} requires only 4.8GB of additional memory for storing the output of the model’s classification head. This is significantly smaller than the dataset itself (>100GB) and can be easily accommodated in the CPU/GPU memory.

Table 10: Comparison of PACE_{fast} memory overhead and the baseline GPU memory requirements.

Dataset	Memory of PACE _{fast}	Baseline GPU Memory	Ratio
CIFAR-100 (VTAB-1K w/ ViT/16-B)	390KB	8.9GB	0.0042%
Camelyon (VTAB-1K w/ Swin-B)	7.81KB	15.7GB	0.000047%
ImageNet (Domain adaptation w/ ViT/16-B)	61MB	8.9GB	0.67%

Resource-Efficient training with PACE_{fast}. Given the superior performance, minimal memory overhead, and no need for additional hyperparameters of PACE_{fast}, we explore its potential for resource-efficient training by maintaining the same number of updates with reduced batch size and proportionally reduced epochs. Table 11 shows that even with 1/8 batch size and epochs, PACE_{fast} still outperforms the baseline by 1.7% while only using $\sim 1/3$ GPU memory and $\sim 1/4$ training time. This demonstrates the robustness and generalization benefits that PACE_{fast} brings to models, enabling them to excel under constrained training configurations. Such an efficiency is particularly valuable for fine-tuning large foundation models, where resource constraints necessitate small batch sizes and typically lead to sharp loss landscapes, yet the theoretical guarantee of PACE for smooth loss landscapes provides a promising solution for these challenges.

Table 11: Results of PACE_{fast} with a reduced batch size and epochs on CIFAR-100 (VTAB-1K w/ ViT-B/16), Camelyon (VTAB-1K w/ Swin-B), ImageNet (Domain adaptaion w/ ViT-B/16). PACE_{fast} outperforms baseline while using less GPU memory and training time.

Method	CIFAR-100			Camelyon			ImageNet			Average		
	Mem.	Time	Acc.	Mem.	Time	Acc.	Mem.	Time	MeanAcc.	Mem.	Time	Acc.
LoRA _{mul} +VPT _{add}	8.9GB	29m	74.6	15.7GB	33m	86.7	8.9GB	161m	44.8	11.1GB	74m	68.7
+PACE _{fast} ($\frac{1}{2}$ batch size, $\frac{1}{2}$ epochs)	5.4GB	17m	78.1	8.6GB	21m	88.9	5.4GB	85m	45.8	6.5GB	41m	70.9
+PACE _{fast} ($\frac{1}{4}$ batch size, $\frac{1}{4}$ epochs)	3.5GB	10m	77.8	6.0GB	14m	88.7	3.5GB	50m	45.6	4.3GB	25m	70.7
+PACE _{fast} ($\frac{1}{8}$ batch size, $\frac{1}{8}$ epochs)	2.9GB	6m	77.2	5.2GB	10m	88.6	2.9GB	32m	45.5	3.7GB	16m	70.4

Table 12: Classification results for different methods on VTAB-1K with different training epochs.

#Epoch	Method	Natural	Specialized	Structured	Avg.
530	GLoRA	83.61	87.02	63.27	77.97
100	Baseline	81.94	85.40	61.40	76.24
100	+PACE	83.94	87.44	64.62	78.67
50	+PACE (half batch size)	83.77	87.32	63.92	78.34
200	Baseline	82.28	85.30	61.64	76.40
200	+PACE	84.13	87.57	64.85	78.85
300	Baseline	82.41	85.00	61.80	76.40
300	+PACE	84.32	87.55	65.13	79.00

D Additional Experiments

In this section, we provide additional experiments of PACE on VTAB-1K with different epochs, varying training data sizes on FGVC benchmarks, self-supervised pre-trained backbones and combinations with other PEFT methods.

D.1 Experiments of VTAB-1K with different epochs

In Table 1, We use 300 epochs for VTAB-1K tasks as we observed slight improvements over 100 epochs. However, this does not mean PACE requires longer training to converge. Since the optimizer uses the cosine learning rate decay, reducing the number of training epochs to 100 has a minimal impact on performance, as shown in Table 12.

To ensure fair memory and computational budgets, we also tested PACE with half the batch size and 50 epochs. Table 12 shows that under these conditions, PACE still improves baseline accuracy by 2.10%, and outperforms the previous SOTA GLoRA, which uses 500 epochs for training and 30 for parameter search. These results demonstrate PACE’s efficiency and effectiveness across various training configurations.

D.2 Experiments on FGVC with limited training data

To validate generalization benefits of PACE on limited data settings, we conduct experiments on FGVC using 50%, 20%, and 10% of the original training samples. Table 13 shows that PACE achieves larger improvements with smaller data sizes, aligning with our theoretical analyses.

Table 13: Classification results on FGVC using varying percentages of data based on ViT-B/16.

Method	CUB			NAB			Flowers			Stanford Dogs			Stanford Cars		
	50%	20%	10%	50%	20%	10%	50%	20%	10%	50%	20%	10%	50%	20%	10%
baseline	87.1	83.9	79.1	80.7	75.0	70.2	98.5	96.5	93.1	90.6	88.7	86.9	78.7	54.9	30.1
+PACE	88.4	85.5	81.4	82.9	77.5	73.8	99.2	97.9	96.1	91.8	90.9	89.8	80.5	57.3	33.2

D.3 Experiments on self-supervised pre-trained backbones

To further verify the effectiveness of PACE on a self-supervised pre-trained backbone, we conduct VTAB-1K experiments on SVHN, Camelyon, and Clevr-Count using MAE [19] and DINO [19], with ViT-B/16 pre-trained on ImageNet-1K [11]. Table 14 shows that PACE improves the baseline on these self-supervised backbones, confirming its applicability to fine-tuning self-supervised models.

Table 14: Classification results on VTAB-1K using self-supervised DINO and MAE, with ViT-B/16 pre-trained on the ImageNet-1K dataset.

Method	MAE			DINO		
	SVHN	Camelyon	Clevr-Count	SVHN	Camelyon	Clevr-Count
Full	90.1	74.6	52.5	89.7	73.1	34.5
Linear	44.5	79.9	57.1	50.7	82.5	44.2
LoRA _{mul} +VPT _{add}	89.3	82.7	82.1	90.0	85.4	55.7
+PACE	93.5	85.8	86.4	91.7	88.1	61.0

D.4 Experiments of Combining PACE with Other PEFT

We conducted experiments combining PACE with several PEFT methods, including AdaptFormer [8], GLoRA [7], COFT [54], and BOFT [42], on CIFAR-100 (VTAB-1K) and ImageNet (domain adaptation) using ViT-B/16. Table 15 shows that integrating PACE improves the baseline performance.

Table 15: Classification results of different PEFT methods based on ViT-B/16.

Method	CIFAR-100 (VTAB-1K)	ImageNet (Domain Adaptation)					
		Source	-Sketch	-V2	-A	-R	Avg.
AdaptFormer	70.6	77.4	26.5	67.4	12.4	28.7	42.4
+PACE	74.8	78.2	27.4	67.9	13.9	31.7	43.8
GLoRA	75.9	78.2	30.3	68.1	13.5	31.6	44.3
+PACE	78.6	78.8	31.7	69.0	15.9	34.4	45.9
COFT	71.8	76.9	26.4	66.7	13.1	30.7	42.7
+PACE	75.3	77.8	27.9	68.2	14.9	32.9	44.3
BOFT	72.3	77.1	27.0	66.8	12.8	31.1	42.9
+PACE	75.7	77.9	28.3	68.2	14.7	33.4	44.5

E Additional Plots

Figures 8 and 9 show the gradient issues in FPA and the gradient regularization effects of PACE.

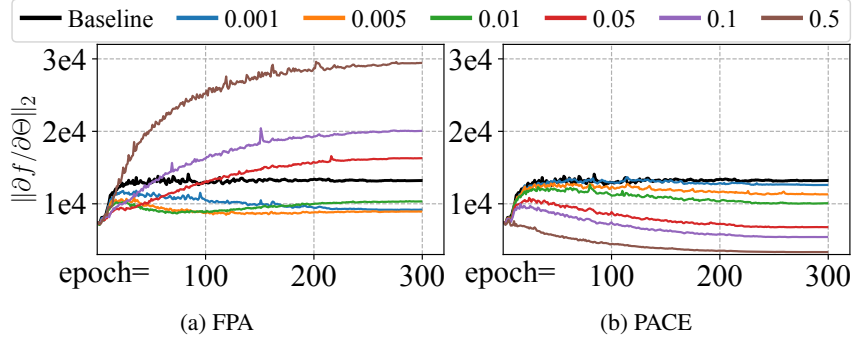


Figure 8: Gradient norms of (a) FPA and (b) PACE with different regularization strengths λ during training on CIFAR-100 (VTAB-1K) w/ ViT-B/16. Figure 5 illustrates the average gradient norm over training epochs.

F Hyperparameter settings

For each dataset, we follow strategies from previous works [41, 28, 7, 45] to apply grid search on the rank, learning rate and weight decay to establish strong baselines. Table 16, 17, 18 and 19 present the hyperparameters and number of trainable parameters used in our strong baseline for VTAB-1K, few-shot learning, FGVC and domain adaptation tasks.

With these strong baselines, we apply grid search on $\lambda \in \{0.02, 0.05, 0.1, 0.2, 0.5, 1\}$ and $\sigma \in \{0.1, 0.5, 1, 1.5, 2\}$ for PACE to optimize its performance.

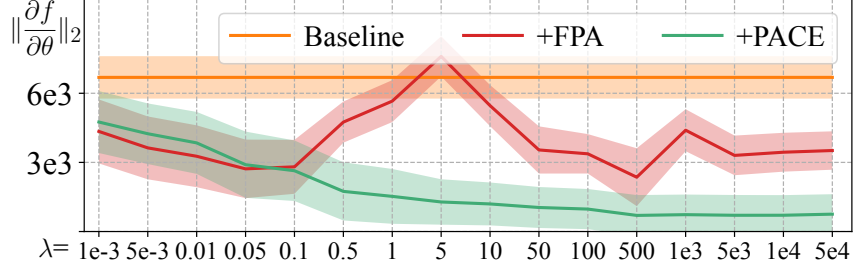


Figure 9: Gradient norms of models across wide range of regularization strengths λ on Camelyon (VTAB-1K) w/ Swin-B. Line and shadow represent mean and std over training epochs. While gradient explosion is less frequent for FPA in this setting, it exhibits unpredictable gradient norm with varied regularization strengths. In contrast, PACE reliably lowers gradient norms as regularization strength λ increases, demonstrating its robustness for effective gradient control.

Table 16: Hyperparameters for baseline on VTAB-1K with ViT-B/16. A: LoRA_{mul}+VPT_{add}, B: LoRA_{add}. lr: learning rate. WD: weight decay.

Hyperparameter	Natural							Specialized				Structured								Average parameter (M)
	Cifar100	Caltech101		DTD	Flowers102	Pets	SVHN	Sun397	Camelyon	EuroSAT	Resisc45	Retinopathy	Clevr-Count	Clevr-Dist	DMLab	KITTI-Dist	dSpr-Loc	dSpr-Ori	sNORB-Azim	
Method	A	A	A	A	A	A	A	A	A	A	B	B	B	A	A	A	A	A	A	B
Rank	10	14	12	18	18	14	10	8	8	10	2	2	8	18	4	10	10	22	4	
lr	1e-3	1e-3	1e-3	1e-3	1e-3	1e-2	1e-3	5e-3	5e-3	5e-3	5e-4	5e-4	1e-4	5e-3	5e-3	5e-3	5e-3	1e-2	2e-4	
WD	1e-4	1e-4	1e-3	1e-2	1e-3	1e-3	1e-2	1e-2	1e-2	1e-2	1e-4	1e-3	1e-4	1e-3	1e-3	1e-4	1e-2	1e-2	1e-2	
1.81																				

Table 17: Ranks for baselines in Few-shot learning. Weight decay is fixed at 1e-4.

learning rate	FGVCAircraft	Food101	Flowers102	OxfordPets	StanfordCars	Mean
Baseline	5e-3	5e-3	5e-3	2e-3	2e-3	Parameter (M)
LoRA _{add}	4	4	4	4	10	0.93
VPT _{add}	1	1	1	1	1	0.14
LoRA _{mul} +VPT _{add}	14	10	18	18	24	2.70

Table 18: Hyperparameters for the baseline LoRA_{mul}+VPT_{add} in FGVC.

Hyperparameter	CUB-200-2011	NABirds	OxfordFlowers	StanfordDogs	StanfordCars	Mean Parameter (M)
learning rate	5e-3	5e-4	5e-3	5e-3	2e-4	2.80
weight decay	1e-2	1e-3	1e-3	1e-2	1e-3	
rank	14	18	18	24	14	

Table 19: Hyperparameters for baseline LoRA_{mul}+VPT_{add} in domain adaptation.

Baseline	rank	learning rate	weight decay	Parameter (M)
LoRA _{mul} +VPT _{add}	10	5e-4	1e-2	2.39

G Experiment details for GSM-8K

We conduct experiments on text generation tasks by fine-tuning Phi-3-mini-4k-instruct [1] on the GSM-8K [9] dataset using causal language modeling. We use learning rate of 2e-6, batch size of 4, LoRA rank of 16, prompt “Answer below question. First think step-by-step and then answer the final number:\n\n<Question>” as instruction and fine-tune models on the training set and evaluated the performance on the test set.

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